# Joint Intent Classification and Slot Filling on ATIS: from Recurrent Networks to BERT

**NLU course project – intent classification and slot filling (lab 5)**

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## Abstract

We train a single model to recognise the intent of a flight-domain request and to label its words with semantic slots. On ATIS, a bidirectional LSTM with dropout raises the slot F1 of an LSTM baseline from 92.24 to 94.59 and its intent accuracy from 93.44% to 94.94%. Fine-tuning BERT, with the slot labels aligned to the first WordPiece of every word, reaches 95.49 slot F1 and 97.54% intent accuracy (means over 5 runs).

## 1. Introduction

Spoken language understanding maps a request to a semantic frame: the intent says what the user wants and the slots carry its arguments. In *list flights from la guardia to burbank* the intent is `flight`, *la guardia* fills `fromloc.airport_name` and *burbank* fills `toloc.city_name`. The two tasks are related and are commonly learnt by one model with two outputs [1]. This report describes the NLU project of the course on ATIS [2]: part A extends the LSTM model of the labs with a bidirectional encoder [3] and dropout [4], part B fine-tunes BERT [5] on both tasks at once [6]. The code is available at `https://github.com/giampyhatesyou/Natural_Language_Processing`.

## 2. Implementation details

**Data and evaluation.** We use the ATIS version of the course labs: 4,978 training and 893 test utterances, 26 intents and 129 slot labels in IOB format [7]. 10% of the training set, stratified by intent, is held out as dev set. Intents are evaluated by accuracy and slots by chunk-level F1 with the conlleval script [8].

**Part A.** Words are embedded in 300 dimensions and encoded by an LSTM with 200 units per direction, which skips the padding through packed sequences. A linear slot layer reads every encoder state and a linear intent layer the final state; in the bidirectional model the intent layer receives the last forward state concatenated with the backward state at the first word, which together summarise the whole utterance. Dropout ($p = 0.3$) is applied to the embeddings and before both output layers. The loss is the sum of the two cross-entropies, minimised with Adam [9] (lr $10^{-3}$, batch size 128, gradients clipped to norm 5).

**Part B.** `bert-base-uncased` feeds a linear intent layer with the pooled `[CLS]` representation and a linear slot layer with the last-layer representation of every sub-token (dropout 0.1). WordPiece [10] splits 2.9% of the test words, but 6.1% of the words inside a slot (city and airport names, aircraft and fare codes), and one test utterance in four contains a split word. Since ATIS labels whole words, the label of a word is assigned to its first sub-token, while the other sub-tokens, `[CLS]`, `[SEP]` and the padding get the ignore index of the loss (Table 2). At inference the prediction of a word is read from its first sub-token, so the slot F1 is computed on the original words with the same script as part A. The pre-trained encoder is loaded once per run and only the two output layers start from scratch. We fine-tune with AdamW [11] (lr $5 \cdot 10^{-5}$, weight decay 0.01 except on biases and LayerNorm), a linear warm-up over the first 10% of the steps followed by a linear decay, batch size 32 and at most 10 epochs.

Table 1: *Test results on ATIS: mean and standard deviation over 5 runs, each evaluated at its best dev epoch.*

| Model | Slot F1 | Intent acc. (%) |
|---|---|---|
| LSTM | 92.24 $\pm$ 0.48 | 93.44 $\pm$ 0.42 |
| BiLSTM | 94.13 $\pm$ 0.41 | 94.54 $\pm$ 0.13 |
| BiLSTM + dropout | 94.59 $\pm$ 0.27 | 94.94 $\pm$ 0.31 |
| BERT | **95.49** $\pm$ 0.10 | **97.54** $\pm$ 0.21 |

**Model selection.** Each configuration is trained 5 times with different seeds. In every run the dev slot F1 is computed after each epoch, training stops after 10 (part A) or 3 (part B) epochs without improvement, and the best weights are restored before testing.

## 3. Results

Table 1 reports mean and standard deviation of the test scores over the five runs. The bidirectional encoder gives the largest gain on slot filling (+1.89 F1), since the backward direction lets the label of a word depend on the words that follow it. Dropout improves both tasks (+0.46 F1, +0.40 points of accuracy) and reduces the spread of the slot F1 across runs. BERT is the best model on both tasks: the margin is large on intents (+2.60 points), where a pre-trained representation of the whole sentence helps most, and smaller on slots (+0.90 F1), which in ATIS largely depend on local lexical cues already captured by the BiLSTM; it is also the most stable model ($\pm$0.10 F1). Averaging over runs matters: the BERT run with the best dev F1, the one kept as checkpoint, has the lowest test F1 of the five (95.33). Thanks to the first-sub-token alignment, the WordPiece segmentation does not penalise BERT: all the models are scored on exactly the same words.

## 4. References

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Table 2: *Training targets of BERT for the test utterance* list flights from la guardia to burbank*: only the first sub-token of each word has a label.*

| Sub-token | Word | Target |
|---|---|---|
| [CLS] | – | ignored |
| list | list | O |
| flights | flights | O |
| from | from | O |
| la | la | B-fromloc.airport_name |
| guard | guardia | I-fromloc.airport_name |
| ##ia | guardia | ignored |
| to | to | O |
| bu | burbank | B-toloc.city_name |
| ##rba | burbank | ignored |
| ##nk | burbank | ignored |
| [SEP] | – | ignored |


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